

MASTER PRODUCTION PROMPT: IMPHERE CIVIC PLATFORM

You are an expert Principal Solutions Architect and Elite Full-Stack MERN Developer. Your task is to generate production-grade, flawless, and optimized code for a civic engagement and social networking platform named "IMPHERE". The platform is commissioned by a local public representative (MLA), meaning it must be highly secure, tamper-proof, scalable, and entirely free of logical flaws before public rollout.

🎨 BRANDING & DESIGN SYSTEM (LUXURY CIVIC)

- **Primary Theme:** Pure, crisp white backgrounds (`#FFFFFF`) and clean off-whites (`#F9F9F9`).
- **Accents:** Polished, rich gold (`#D4AF37` / `#C5A059`) used exclusively for active states, high-priority interactive components, icons, badges, and primary action buttons.
- **Typography:** Premium serif headings paired with clean, highly readable geometric sans-serif body text.
- **Responsiveness:** Fluid grid and flexbox layouts utilizing Tailwind CSS to adapt flawlessly between mobile screens and desktop monitors.

🛠️ TECH STACK & ARCHITECTURAL REQUIREMENT

- **Frontend:** React.js, Tailwind CSS, Lucide React (for icons), and Vite.
- **Backend:** Node.js, Express.js implementing a modular MVC directory architecture.
- **Database:** MongoDB with robust Mongoose schemas, indexing, and validation.
- **Authentication:** Google OAuth 2.0 (JWT-based session management, HttpOnly cookies).
- **Anti-Cheat Validation:** HTML5 Geolocation API on the frontend paired with server-side photo metadata inspection (using `exifreader`). Uploading images from the device gallery is strictly prohibited; all proof must be captured live.
- **Content Moderation:** Integration placeholders for an automated image/video moderation API to strictly block explicit or 18+ content.

📄 DATA SCHEMA REQUIREMENTS

Implement the following core schemas with strict types and validations:

1. **User Schema:** Stores account details, role (`'user'`, `'admin'`), `activePin` (current GPS location), `nativePin` (user-selected hometown, changeable only once a year), `standing` (non-spendable civic reputation score), `impactCredits` (spendable reward currency), `badge` tier (`'Citizen'`, `'Bronze'`, `'Silver'`, `'Gold'`), and tracking arrays for `believers` (followers) and `believing` (following).
2. **Challenge/Proclamation Schema:** Tracks both pre-defined static tracks (Fitness, Tree Plantation, Blood Donation) and user-raised Proclamations. Tracks geolocation coordinates, an

array of `backers`, a `powerThreshold` configuration, and distinct rewards for both `standingReward` and `icReward`.

3. **Voucher Inventory Schema:** Securely holds pre-purchased, encrypted alphanumeric merchant codes/vouchers that cannot be converted back into real fiat currency.

📱 PAGE-BY-PAGE COMPLETE UI/UX & LOGIC CODE GEN

Generate complete, clean, modular component structures and matching API endpoints for these 9 core screens:

1. **Authentication & Onboarding:** Split-screen layout on desktop, vertical stack on mobile. Includes Google Sign-In and a mandatory first-time onboarding modal to lock in the user's "Native Pin".
2. **Home Page (The Civic Feed):** Fixed navigation containing a 3-tab segmented selector block: `[ For You ]` (moderated global feed), `[ Believing ]` (chronological following feed), and `[ Challenges ]` (public actions). Post cards must feature elegant thin borders and gold outline icons for Vouch (Like), Comment, Share, and Save.
3. **Search & Explore:** Rounded search bar leading to a masonry grid layout displaying trending regional hashtags and suggested citizen profiles.
4. **Challenges Hub:** Split-screen on desktop, vertically stacked on mobile. Displays a locality dropdown menu.
 - **Tab A (Welfare Tracks):** Lists static challenges with solid gold "Accept" buttons.
 - **Tab B (Civic Proclamations):** Contains a "Raise an Issue" dash-bordered trigger that opens a live-camera-only submission form. Displays active crowd-backed issues showing a progress metric tracking votes against the `powerThreshold`.
5. **Community Challenges (Impact Circles):** Features a tier-gated lock screen for accounts below the Bronze Badge. For Bronze and above, it provides an Impact Circle dashboard with a "Ready to Serve" live worker availability roster, active group missions, and an Eminence-based team leaderboard.
6. **The Exchange (Shop Page):** Displays a prominent gold wallet card showing the spendable Impact Credits (IC) balance. Renders a grid of voucher cards that, when purchased, execute backend balance validation and display a secure alphanumeric voucher with an auto-generated QR code.
7. **Notification Center:** A crisp single-column vertical list with small gold dot indicators for unread entries, sorting social engagements from critical civic alert updates.
8. **Profile (Civic Resume):** Features a bold centered avatar framed dynamically by the user's current metallic tier color. Displays structural counts for Believers, Believing, and total Standing. Features a dual-grid portfolio displaying regular posts alongside verified, gold-shield watermarked historical challenge completions.
9. **Settings Screen:** Provides clean, row-by-row configurations for Account Password adjustments, Profile Data editing, and Interface layout preferences.

🚀 EXECUTION INSTRUCTION

Provide the production-ready code step-by-step. Start by providing the backend architectural setup (server, database connections, and schemas), followed by the API routing controllers, and finally the frontend React components complete with Tailwind utility classes. Avoid placeholders or truncated code sections. Keep all logic explicit, robust, and completely secure.

To make this platform highly prestigious and flawless for public launch, we will establish a **"Luxury Civic" visual language**. The theme utilizes a pure white and off-white background system paired with rich, polished gold accents (#D4AF37 / #C5A059) for high-priority interactive components, icons, badges, and primary call-to-actions.

Here is the exhaustive UI/UX wireframe blueprint and design spec for every single page of the **Imphere** platform, optimized for a fluid layout that adapts flawlessly between mobile devices and desktop viewports.

🗝️ 1. Authentication & Onboarding Stage

Desktop Layout

A balanced 50/50 split-screen layout.

- Left Panel (50% width):** Deep matte black background featuring a large, glowing gold geometric icon of the Imphere emblem. Below the logo, the motto reads: **"Build Your Standing. Resolve the Future."** in elegant serif typography.

- Right Panel (50% width):** Clean, crisp white background hosting the interaction block.

Mobile Layout

A singular, vertically centered stack on a solid white background. The gold Imphere logo sits proudly at the top center, with the interaction block structured cleanly beneath it.

Core Interface Components

- Role Selection Block:** A sleek, minimal horizontal segmented control with a micro-interaction slide effect: [Citizen Portal] (Default) and [Admin Console].

- Google Sign-In Button:** A premium, wide pill-button featuring a crisp white background, a thin, solid gold border, and the official Google "G" icon centered next to the label: "Continue with Google".

- Onboarding Overlay (First-Time Sign-In Only):** A modal overlay that prompts the user to configure their geographic identity:

- A descriptive helper text explaining the **"Dual-Pin System"**.

- Native Pin Input:** A searchable geolocation dropdown menu where users firmly select their home municipality or town. Once saved, a layout block specifies: **"Your Native Pin can only be modified once per year."**

🏠 2. Home Page (The Civic Feed)

Navigation Architecture

- Desktop:** A fixed, left-hand vertical sidebar containing clean text labels alongside functional icons. The top of the sidebar displays a prominent gold "IMPHERE" wordmark.

* **Mobile:** A fixed, ultra-clean top app bar featuring the gold "IMPHERE" logo on the left and a notification bell icon on the right. A sticky bottom tab navigation bar handles core view switching.

Feed Switcher (The Segmented Header)

Directly beneath the header sits a premium, three-column tab bar with a subtle gold accent line indicating the active selection:

1. **[For You]:** A fluid vertical scroll showing public engagement media.
2. **[Believing]:** A focused timeline displaying chronological posts exclusively from Catalysts they follow.
3. **[Challenges]:** A specialized feed detailing community-driven actions and completed civic tasks.

The Core Post Component Card

A modern, card-based layout featuring a crisp white surface bordered by a razor-thin line (#E5E5E5).

* **Header Area:** Displays the Catalyst's circular avatar image, framed nicely by a thin colored border representing their earned badge tier (Bronze, Silver, Gold). Next to the avatar, the user's name is displayed above a small, faded geographic location indicator tag.

* **Media Frame:** An aspect-locked display block (supporting high-resolution images or video clips). ***Backend Logic Safeguard:** Any upload hitting this viewport passes through an integrated validation layer to catch and filter out any inappropriate or adult media.

* **Interaction Toolbar:** Positioned cleanly below the media asset. Features four distinct, high-contrast gold outline icons that transition to solid gold when selected:

* **Vouch (Like):** Represented by an elegant chevron or thumbs-up icon.

* **Comment:** A clean text-bubble icon.

* **Share:** A minimalist paper-airplane or node icon.

* **Save:** A sleek ribbon bookmark icon.

🔍 3. Search & Explore Page

Search Header

A full-width, premium search bar element. It features a soft off-white background container rounded tightly at the corners, decorated with a solid gold magnifying glass icon on the left edge. The placeholder text reads safely: "Search Catalysts, Impact Circles, or local tags...".

Layout Grid Architecture

* **Desktop:** A beautifully balanced, 4-column masonry layout block.

* **Mobile:** A responsive, alternating 2-column staggered image and video grid.

Grid Content Cards

* **Trending Civic Stories:** Image blocks featuring a dark gradient overlay on the bottom edge, overlaying crisp white text highlighting trending regional hashtags (e.g., #GudurGreenDrive).

* **Suggested Catalysts:** Small, rounded profile square cards displaying an avatar, username, total **Standing Score**, and a clear gold button labeled "Believe".

🏆 4. Challenges Page (The Heart of Imphere)

Locality Selection Hub

A stylized header card component wrapped in a thin gold border. It features a dropdown menu displaying the user's current active geographic location based on their live device coordinates. A

contextual info button reminds users that switching this tab changes their active view to display localized issues and challenges.

Interface Splitting (Dual Component Tabs)

Tab A: Welfare Tracks (Static Challenges)

An organized collection of vertical list cards representing pre-packaged civic paths:

- * **Card Archetypes:** *Fitness Tracks, Blood Donation, Food Donation, Tree Planting, Infrastructure Maintenance.*

- * **Card Interior:** Decorated with a bold gold category icon on the left. The right side features a bold task title (e.g., "Water 5 Public Trees"), a horizontal progress metric tracking community completions, and a clear point breakdown indicator: +50 Standing (Impact Score) | +20 Impact Credits (IC).

- * **Action Button:** A solid gold button block labeled "Accept Challenge".

Tab B: The Civic Proclamation Engine (User-Raised Issues)

A highly organized feed built directly around active user-reported entries.

- * **The "Raise an Issue" Trigger:** A prominent, full-width white banner card with a solid gold dashed border. It features an inviting text string: "Spot a local problem? Raise a Proclamation." Clicking this component opens a full modal screen:

- * **Form Fields:** Problem Title, Description input field, and a Category dropdown selector.

- * **Anti-Cheat Camera Module:** A dedicated camera frame component. The standard system "Choose file from gallery" option is completely omitted. It features a large gold circular capture button. When triggered, the interface locks the precise device coordinates via the browser's location API to ensure absolute authenticity.

- * **The Voting Ballot Cards:** Displays active local problems awaiting consensus. Each card showcases the user-submitted photo, an active countdown timer tracking the 14-day resolution window, and a highly visible vote collection bar:

- * **Progress Bar:** Visually tracks the current vote tally against the item's established **Power Threshold** (e.g., 342 / 500 Backers).

- * **Action Call:** A wide button labeled "Back This Issue". Selecting it adds the user's personal **Standing Score** to the collective weight of the mandate.

5. Community Challenges Page

Tier-Gated Creation Banner

- * **Unlocking Logic:** If a user's current profile tier resides strictly below the **Bronze Badge** milestone, this area displays an informative, greyed-out dashboard component decorated with a secure lock icon and an advisory string: "Achieve Bronze Badge rank to unlock Impact Circle creation."

- * **Active State:** For users meeting the rank requirement, this banner transforms into a striking, gold-bordered action block displaying a button labeled "Form Impact Circle". Clicking this launches a configuration screen to define the group's name, upload an identity avatar, and write their core mission statement.

Group Dashboard Layout

A dual-column administrative interface on desktop viewports, collapsing cleanly into a unified tabbed system on mobile.

- * **The "Ready to Serve" Hub:** A dedicated control panel card for the community manager (Principal/Steward). It features a live grid displaying avatars of all active members who have

currently toggled their availability switch to "On". This allows managers to seamlessly delegate tasks without broadcasting notifications to the entire group.

- * **Active Roster Panel:** Displays massive civic challenges currently undertaken by the group (e.g., "Town Park Restoration Drive").

- * Includes a real-time distribution control module accessible to the community leader, featuring input fields to cleanly divide and award earned Impact Credits directly out to participating members' accounts once a task is successfully resolved.

- * **Eminence Leaderboard Tab:** A clean, tabular list ranking regional Impact Circles based entirely on their collective **Eminence Score**.

🛒 6. The Exchange (The Shop Page)

Balance Summary Card

A beautifully styled, solid gold rectangular card that sits prominently at the top of the viewport. It displays the user's available, spendable currency balance in a large serif font: 1,450 IC, accompanied by a descriptive label reading: "Impact Credits Balance (Not convertible to fiat currency)".

Reward Catalog Grid

A highly structured multi-column display card layout.

- * **Product Card Anatomy:**

- * **Top Half:** A clean product display box containing images of reward items (e.g., *Digital Learning Materials, Local Vendor Gift Cards, Fitness Passes*).

- * **Bottom Half:** Displays the item title alongside a prominent gold pill tag indicating its exact purchase price (e.g., 500 IC).

- * **The Redemption Flow:** Clicking a product card opens a focused confirmation dialog box. Upon confirming the purchase, the backend handles transaction safety checks, deducts the credit amount from the database, and renders a secure modal displaying a unique, high-contrast alphanumeric voucher string along with an auto-generated QR code for real-world merchant redemption.

🔔 7. Notification Page

Layout Architecture

A clean, single-column vertical list layout restricted to a maximum readable width of 800px on desktop viewports to ensure high legibility.

Notification Item Components

Every item is presented inside a thin, horizontally split row container. Unread notifications are easily identified by a small, solid gold circular badge indicator on the left border edge.

- * **Social Interactions:** Displays standard text strings paired with interactive profile links (e.g., "[User A] and 3 others vouched for your civic post.>").

- * **Civic System Alerts:** High-contrast notification items detailing critical structural updates, styled with specific alert formatting:

- *  **"Action Required: A Proclamation you backed in your hometown has reached its Power Threshold and transitioned into an active challenge!"**

- *  **"Success: Your Proof of Action for 'Tree Plantation' has been fully verified. +20 IC credited."**

👤 8. Profile Page (The Civic Resume)

Identity Header Block

* **Avatar Assembly:** A large, high-resolution circular profile photo centered in the viewport. The avatar is framed by a thick profile ring whose metallic color dynamically reflects the user's current tier achievement level (Bronze, Silver, or Gold).

* **Metrics Dashboard:** A horizontal row containing three high-visibility data blocks:

* **[1,205] Believers** (Followers)

* **[432] Believing** (Following)

* **[8,450] Standing** (The user's permanent, non-spendable reputation metric).

The Civic Portfolio Grid

A multi-tabbed lower layout structure:

* **Tab 1: Media Feed:** Displays a standard three-column square thumbnail grid of all regular lifestyle and civic images shared by the account, matching modern social layouts.

* **Tab 2: Verified Actions Portfolio:** A specialized, premium grid display showcasing card summaries of every single verified challenge the user has successfully executed. Each thumbnail is permanently watermarked with a small gold shield icon in the top corner, serving as an unalterable, cryptographically secure digital resume of their real-world civic impact.

9. Settings Page

Layout Configuration

A split layout on desktop screens featuring a left-hand navigation list and a spacious right-hand interaction panel. On mobile viewports, it loads as a clean vertical settings tree.

Core Configuration Options

Every settings category is built into an expandable menu row container with a clean white background and sharp text labels:

* **Account Credentials Security:** Opens a secure form interface featuring validation inputs to update user account passwords safely.

* **Profile Customization Hub:** Provides standard input fields to modify public account details, update display text, or replace avatar graphics.

* **Interface Preferences:** Features a dedicated, toggleable theme control block to customize operational views across available system layout modes.

Final Architecture Confirmation

This comprehensive wireframe design spec successfully balances a high-end social aesthetic with robust, tamper-proof civic validation workflows. Every single view is designed from the ground up to ensure high public trust and prevent point manipulation.

To ensure **Imphere** is built to a flawless, production-grade standard that can withstand public scrutiny and security audits, every technology in your MERN-based stack must serve a specific, strategic purpose.

Here is the comprehensive, deep-dive detail of every single technology, framework, and module required to power the application securely and smoothly.

Frontend Tier (The Client Application)

1. React.js (via Vite)

* **What it does:** The core structural framework for building your user interface using reusable, state-driven components.

* **Production Role:** Instead of using older build tools like Create React App, **Vite** is utilized to compile the frontend. Vite uses native browser ES modules, giving your development environment near-instant loading times and creating highly optimized, minified production builds that load lightning-fast on budget mobile devices.

* **Key Modules to Install:** npm install vite react react-dom

2. Tailwind CSS & Shadcn/ui

* **What it does:** Tailwind is a utility-first CSS framework for rapid UI styling directly inside your HTML markup. Shadcn/ui is a collection of beautifully designed, accessible components built on top of Tailwind.

* **Production Role:** This combination ensures your **White & Gold luxury civic theme** looks identical and renders smoothly across both desktop viewports and mobile phone browsers. Shadcn components (like modals for submitting challenges or dropdowns for selecting your native pin) are pre-engineered for absolute structural stability and perfect mobile responsiveness out of the box.

* **Key Modules to Install:** npm install tailwindcss postcss autoprefixer lucide-react

3. TanStack Query (React Query)

* **What it does:** An advanced data-fetching, caching, and state synchronization library for React.

* **Production Role:** In an app where citizens are constantly looking at live public feeds, voting on local problems, and tracking leaderboards, standard state management (useState/useEffect) will quickly bottle-neck your app. React Query handles automatic background refetching (keeping vote tallies synchronized without manually refreshing the screen) and provides **Optimistic Updates**—when a user clicks "Vouch" or "Back This Issue", the UI changes instantly while the network request resolves in the background, making the app feel incredibly snappy.

* **Key Modules to Install:** npm install @tanstack/react-query

Backend Tier (The Logic Server)

4. Node.js & Express.js (With TypeScript)

* **What it does:** Node.js is the asynchronous, event-driven JavaScript runtime environment. Express is the minimal, flexible web framework that handles your server routing, middleware, and API endpoints.

* **Production Role:** Compiling your server using **TypeScript** is a non-negotiable requirement for an application backed by a public representative. TypeScript forces strict type safety across your codebase. This eliminates common runtime bugs, ensures that data flowing into your reward wallets or database is perfectly formatted, and makes your backend modular, clean, and easily auditable.

* **Key Modules to Install:** npm install express typescript @types/node @types/express ts-node-dev

5. JWT (JSON Web Tokens) & Cookie-Parser

* **What it does:** Handles secure, stateless session management for authenticated users.

* **Production Role:** Once a user signs in via Google, the backend issues an encrypted JSON Web Token. For production security, this token is *never* stored in the browser's localStorage (where it can be stolen by malicious scripts). Instead, it is transmitted via an **HttpOnly, Secure, SameSite Cookie**, shielding user sessions from Cross-Site Scripting (XSS) attacks.

* **Key Modules to Install:** npm install jsonwebtoken cookie-parser

🗄 Database & Caching Tier (The Storage Layer)

6. MongoDB & Mongoose

* **What it does:** MongoDB is a document-based, NoSQL database that stores data natively as flexible, JSON-like objects. Mongoose is the Object Data Modeling (ODM) library that manages business logic rules, data validation, and relations on top of MongoDB.

* **Production Role:** MongoDB is ideal for social and civic feeds because its dynamic schema allows user posts, completed challenge media, and diverse static welfare categories to scale without rigid database schema migrations. Mongoose maps strict data rules over your collections, ensuring that a user's permanent **Standing Score** cannot be accidentally updated by a faulty challenge router.

* **Key Modules to Install:** npm install mongoose

7. Redis

* **What it does:** An ultra-fast, in-memory, key-value data structure store used primarily as a cache and message broker.

* **Production Role:** **Crucial for your system leaderboards and feeds.** If thousands of citizens are actively loading the local leaderboards or sorting the "For You" feed, querying millions of rows across MongoDB directly will cause database locks and crash your server. Redis holds the top-ranked individual and community scores in RAM, serving leaderboard positions to users in milliseconds with virtually zero processing overhead.

* **Key Modules to Install:** npm install redis

🛡 Security, Anti-Cheat, & Infrastructure Tier

8. HTML5 MediaDevices & Geolocation API

* **What it does:** Native web APIs that grant developers secure access to hardware features like the physical mobile camera and raw GPS coordinate data.

* **Production Role:** To enforce the **"In-App Eye" workflow**, the frontend omits the traditional `<input type="file">` file-picker, meaning users cannot tap their system photo gallery. Instead, the `MediaDevices.getUserMedia()` API opens a secure, sandboxed camera viewfinder directly inside the application, forcing citizens to snapshot their Proof of Action live. Simultaneously, the Geolocation API captures their precise coordinates at the split-second the shutter closes.

9. ExifReader & Geolib (Backend Counter-Measures)

* **What it does:** `exifreader` parses deep binary metadata embedded directly into image files (EXIF data). `geolib` runs precise geospatial distance calculations between pairs of coordinates.

* **Production Role:** **The absolute core of your anti-cheat pipeline.** If a malicious user attempts to bypass your frontend constraints by simulating an image upload directly to your backend API via Postman, the backend uses `exifreader` to extract the hidden camera metadata from the raw photo. It extracts the physical time and true coordinates embedded in the image and passes them to `geolib`. If the distance between where the user *claims* they are and where the photo *proves* they are is greater than 50 meters, the transaction is flagged, and the upload is safely blocked.

* **Key Modules to Install:** npm install exifreader geolib

10. AWS Rekognition API (Automated Moderation)

* **What it does:** A cloud-based, deep-learning computer vision service that automatically detects explicit, inappropriate, or unsafe content within images and videos.

* **Production Role:** Because this platform is publicly marketed and sponsored by an MLA, you cannot afford to have explicit or 18+ content sitting on your "For You" community feed before human moderators can look at it. The moment an image or video bypasses your anti-cheat verification, the backend instantly dispatches the media stream to AWS Rekognition. If the API returns an unsafe confidence flag above a strict threshold, the media is automatically expunged, and the user's workflow is halted.

11. Cloudinary / AWS S3 (Media Storage CDN)

* **What it does:** Cloud-based object storage services designed for hosting and globally distributing heavy static media assets (images, audio, video).

* **Production Role:** Storing image and video uploads directly inside your database or on your Node.js server storage will rapidly exhaust your resources and bottleneck network speed. Instead, media is seamlessly routed to an external cloud bucket. Using dynamic transformation settings (e.g., via Cloudinary), uploaded images are automatically compressed, cropped, resized, and converted to modern formats like .webp on the fly, reducing asset sizes by up to **70%** and keeping feed load times remarkably low.

* **Key Modules to Install:** npm install cloudinary (or @aws-sdk/client-s3)

12. Crypto Node Module (Voucher Encryption)

* **What it does:** A built-in, native cryptographic utility package provided by Node.js for data encryption, decryption, and secure hashing.

* **Production Role:** The e-commerce vault stores real, unredeemed merchant voucher codes and digital rewards. If your database is ever compromised, plain-text coupon strings could be exposed and cleaned out. The backend uses the production-grade **AES-256-GCM encryption algorithm** to securely cipher voucher strings before they are written to disk. The codes are decrypted in-memory only at the exact millisecond a valid user successfully redeems their Impact Credits, ensuring absolute security for sponsor assets.

Summary Configuration Chart

Stack Component	Chosen Tool	Critical Security Setting
Frontend Runtime	Vite + React	Secure Environment Separation (.env)
Backend Framework	Node.js + Express (TypeScript)	Strict Type Validation Middleware
Primary State Handler	TanStack Query	Automated Data Staling & Smart Refetching
Main Database	MongoDB + Mongoose	Read/Write Compound Indexes on Geolocation
Performance Layer	Redis	Least-Recently Used (LRU) Cache Eviction
Session Control	JWT	HttpOnly, Secure, SameSite=Strict Cookies
Media Pipeline	Cloudinary / AWS S3	Authenticated Signed Upload Signatures
AI Moderation Layer	AWS Rekognition	Preserved Strict Confidence-Score Rejection
Verification Math	ExifReader + Geolib	Deep Hexadecimal Metadata Check

To launch **Imphere** quickly under your local MLA's sponsorship while ensuring the platform is completely secure, bug-free, and fraud-resistant, you need a highly focused **Minimum Viable Product (MVP)** feature blueprint.

Below is the definitive, production-ready specification of the absolute minimum functionality, data requirements, and backend logic required for every single page. This blueprint strips away non-essential clutter and focuses entirely on the core engine so you can build a flawless, un-hackable application.

🗝️ 1. Authentication & Onboarding Stage

The MVP goal is to establish secure user identities and lock down their geographic boundaries right at registration.

Core Functions & Logic

- * **Google OAuth 2.0 Integration:** Users authenticate via Google. The backend checks if the email exists. If yes, it signs them in and issues an HttpOnly JWT cookie. If no, it creates a new user record and flags their profile state as `onboarding_incomplete`.

- * **Hometown Anchor Lock:** Users with `onboarding_incomplete` are immediately intercepted by a full-screen block. They must select their home village or town from a secure dropdown menu. Once selected, the backend saves this as their permanent `nativePin` and updates their status to `active`.

MVP Database Payload (Strict Minimum)

```
``json
{
  "email": "string (unique)",
  "role": "string ('user' | 'admin')",
  "nativePin": "string (hometown name/code)",
  "onboardingStatus": "string ('incomplete' | 'active')"
}
```

...

📱 2. Home Page (The Civic Feed)

The MVP goal is to deliver a reliable, clean layout split into three distinct, high-performance feeds.

Core Functions & Logic

- * **Feed Segmentation Switcher:** A simple, responsive three-tab header:

1. **For You:** Displays a global feed of verified civic actions. The backend must include a basic filter checking a boolean `isApproved` flag to prevent unmoderated or explicit posts from appearing.

2. **Believing:** Displays posts from users found within the current user's believing array.

3. **Challenges:** Displays posts explicitly tagged as completed welfare challenges.

- * **The MVP Post Component:** A clean, vertical card listing user info, the media file, and 4 functional buttons (Vouch, Comment, Share, Save). Clicking **Vouch** increments a numerical counter in MongoDB instantly and changes the icon to solid gold.

🔍 3. Search & Explore Page

The MVP goal is to allow citizens to discover other active users and trending local initiatives.

Core Functions & Logic

* **Plain-Text Search Endpoint:** A single search input field that triggers a backend regex database query against user names and challenge hashtags.

* **Dynamic Discovery Cards:** Renders two simple sections below the search bar:

* **Top Local Catalysts:** A small grid displaying active local users sorted by their total standing score.

* **Trending Content:** A grid showing recent highly-vouched post thumbnails.

🏆 4. Challenges Page (The Core Civic Engine)

This is the most critical part of your production engine. The MVP goal is to present pre-defined tasks while allowing trusted local community members to escalate real-world problems into funded challenges.

Core Functions & Logic

* **Static Track Selector:** A list component rendering core welfare categories: *Fitness, Blood Donation, Tree Plantation, Food Donation*. Clicking a category opens a list of pre-defined tasks.

* **The Anti-Cheat Cam Submission Form:** Clicking "Submit Proof" opens a specialized frontend modal overlay:

1. The standard `<input type="file">` is removed. The frontend uses the browser's native `navigator.mediaDevices.getUserMedia` API to stream the live phone camera into a viewfinder element.

2. When the user snaps the photo, the frontend requests their current position via `navigator.geolocation.getCurrentPosition`.

3. The image file and raw latitude/longitude coordinates are sent together as a single multipart/form-data payload to the server.

4. **Backend Telemetry Check:** The Node.js server extracts the hidden location data inside the image file using the `exifreader` module. If the coordinates from the phone's GPS do not match the embedded image metadata within a 50-meter radius, the submission is rejected as a location-spoofing attempt.

* **The Proclamation (Civic Mandate) Engine:** * Any citizen can click "Raise Proclamation" to report a local problem (e.g., a broken water pipe), which costs them 5 non-spendable standing points to prevent spam.

* The problem is listed as a local poll. Other local users can click "Back This Issue".

* Once the total accumulated standing scores of all backers crosses the problem's predefined `powerThreshold`, the status shifts automatically to a **Live Local Challenge**.

MVP Database Payload (Strict Minimum)

```
``json
{
  "title": "string",
  "type": "string ('static' | 'proclamation')",
  "status": "string ('pending' | 'active' | 'completed')",
  "coordinates": { "lat": "number", "lng": "number" },
  "backers": ["ObjectId (User)"],
  "powerThreshold": "number",
  "standingReward": "number",
  "icReward": "number"
}
```

...

🧑‍🤝‍🧑 5. Community Challenges Page

The MVP goal is to let high-performing citizens organize groups to execute large-scale public tasks collectively.

Core Functions & Logic

* **Tier-Gated Creation Logic:** The backend checks the user's badge string. If the value is strictly equal to "Citizen", the "Create Community" button is disabled and displays a message: "Achieve Bronze Badge rank by completing individual challenges to unlock."

* **The "Ready to Serve" Roster:** Inside an Impact Circle page, members can toggle a simple switch component. This sets their database boolean readyToServe to true. The community leader (Principal) can see an active roster list of all available members in their locality to easily assign group tasks.

* **Eminence Rankings:** A basic tabular table that groups all users by their community ID, sums their total collective reputation, and ranks the communities on a local leaderboard.

🛍️ 6. The Exchange (The Shop Page)

The MVP goal is to let users redeem their hard-earned points for digital codes without allowing conversions back into physical fiat cash.

Core Functions & Logic

* **Live Balance Display Tracker:** A prominent container displaying the logged-in user's spendable impactCredits balance.

* **Atomic Voucher Redemption Pipeline:** When a user clicks "Redeem" on a voucher card (e.g., a ₹100 Local Store Coupon costing 500 IC):

1. The backend runs a transactional query to check if the user's impactCredits balance is greater than or equal to 500.
2. It deducts exactly 500 IC from the user's account.
3. It fetches a single unredeemed coupon string from your encrypted Vouchers database collection.
4. It marks that specific voucher document as isRedeemed: true and links it to the user's ID.
5. The frontend displays the unique alphanumeric voucher code on screen alongside a matching QR code generated via a basic frontend library like qrcode.react.

MVP Database Payload (Strict Minimum)

```
``json
{
  "couponCode": "string (encrypted via AES-256)",
  "valueDescription": "string",
  "icCost": "number",
  "isRedeemed": "boolean",
  "redeemedBy": "ObjectId (User)"
}
```

...

🔔 7. Notification Page

The MVP goal is to alert users about social interactions and critical local updates.

Core Functions & Logic

- * **Unified Activity List:** A vertical list mapping rows from a Notifications database collection.
- * **Civic Escalation Alerts:** The system auto-generates notifications when:
 - * Another user comments or vouches for their post.
 - * A local issue they chose to back successfully hits its threshold and turns into an active public challenge.

👤 8. Profile Page (The Civic Resume)

The MVP goal is to showcase a citizen's verifiable real-world achievements to build trust and career reputation.

Core Functions & Logic

- * **Dynamic Tier Ring:** Renders the user's circular avatar image wrapped in a colored CSS border matching their database badge value:
 - * Bronze: #CD7F32 | Silver: #C0C0C0 | Gold: #D4AF37
- * **The Unalterable Civic Resume Grid:** A two-tab portfolio display:
 - * **Grid 1 (Social):** Displays standard everyday user posts and content.
 - * **Grid 2 (Verified Actions):** Automatically queries and displays only the challenge submissions where the backend status is strictly marked as completed. Every thumbnail card in this tab is overlaid with a solid gold lock or shield icon to signify its authenticity.

⚙️ 9. Settings Page

The MVP goal is to give users direct control over their basic operational preferences and security details.

Core Functions & Logic

- * **Profile Editor Form:** Input fields allowing users to modify their display name and profile picture URL.
- * **Security Update Handler:** A secure form endpoint that accepts old and new passwords, hashes the new entry using bcrypt, and updates the database record (applicable only to traditional fallback accounts or administrators).

🎯 Next Steps for Your Development Team

This specification outlines the essential core logic needed for a secure, functional production launch. You can hand this clear, logical structural breakdown directly to your engineering team to set up your MongoDB collections, write your Node.js controllers, and implement your React components systematically.

Would you like to review the specific database indexing parameters for the geospatial queries next, or should we focus on structuring the authentication middleware code for your backend?